Student number: ______________________________

University of Cape Town

Department of Electrical Engineering

EEE3094S 2025: Class Test 1

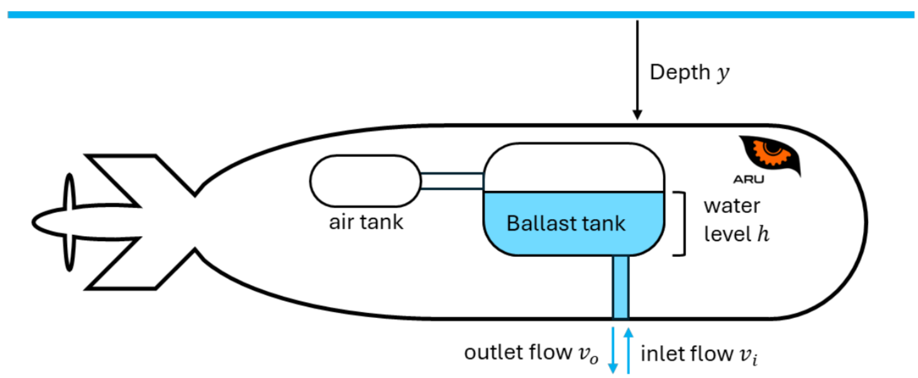
Test Information

- The test is open book. You do not need to hand in your resources.
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write "see attached pages" by the question) and make sure they are marked with your student number.
- You may do your diagrams in pencil but please ensure that your lines are clear.
- There are 3 questions worth a total of 50 marks.
- You have 90 minutes to complete the test + 5 minutes of reading time.

Question 1 [15 marks]

Marine robots, or autonomous underwater vehicles (AUVs) are a promising technology for exploring the depths of the oceans, where it is difficult and dangerous to send humans. Let's say at some point in the future, the ARU's work in Southern ocean research leads to a new project designing an AUV to study sea life in the depths below the Antarctic ice field, and you are on the team. Your job: depth control.

Figure 1: Depth control system in the AUV



Information:

Like a submarine, the AUV will control its depth using the balance of air and water in ballast tanks. When the AUV wants to dive downwards, the ballast tank lets in water, making the robot become negatively buoyant so it sinks. When it wants to go back up, compressed air stored aboard the robot is pumped into the ballast tank and water pushed out, making the robot become positively buoyant and float.

The water level $h(s)$ affects the dynamics of the AUV, $G(s)$, causing its depth $y(s)$ to change. The depth is measured using the depth sensor $H_1(s)$, and compared to the reference $y_r(s)$. The resulting error is given as input to the depth controller $C_1(s)$, which selects the desired water level for the ballast tank, $h_r(s)$.

The actual level of water in the ballast tank, $h(s)$, is measured by a level sensor $H_2(s)$. The difference between the reference and measured level ($e(s)$) is acted upon by the level control system. This consists of two valve control systems. If the error in the water level is positive or zero (that is, the water level needs to increase), the inlet valve controller $C_i(s)$ opens the inlet valve $V_i(s)$, allowing water to come through with inlet flow rate $v_i(s)$. If the error in the water level is negative (needs to decrease), the outlet valve controller $C_o(s)$ opens the outlet valves $V_o(s)$, letting compressed air into the tank and allowing water to be pushed out with outlet flow rate $v_o(s)$.

The total flow rate into the ballast tank is then the outflow rate subtracted from the inflow rate. (Note that, in reality, water will never flow in both directions at the same time.) The geometry of the ballast tank $B(s)$ transforms the flow rate into the rate of change in the water level $\dot{h}(s)$, the integral of which is the level $h(s)$, that is read by the sensor.

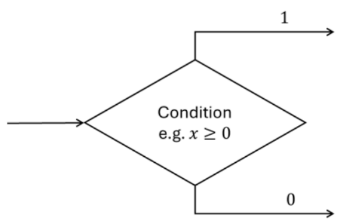
Questions:

a) Create a block diagram representing depth control of the AUV. [10 marks]

b) Reduce your block diagram to derive a transfer function model relating the reference $y_r(s)$ and the depth $y(s)$ for the case where the AUV is diving down, in terms of the given subsystem labels. [5 marks]

Hints:

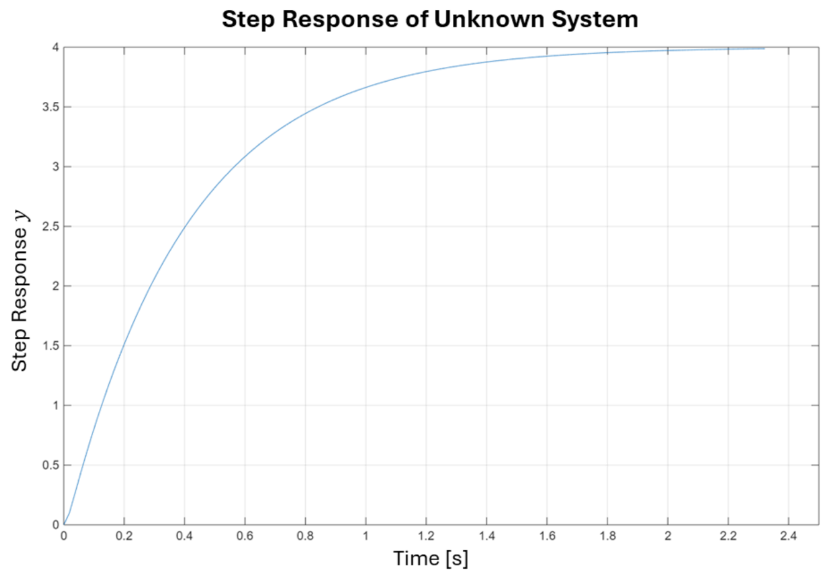
- There is an IF statement in this model. You may represent this in the block diagram as you would in a flow chart (see the example below).



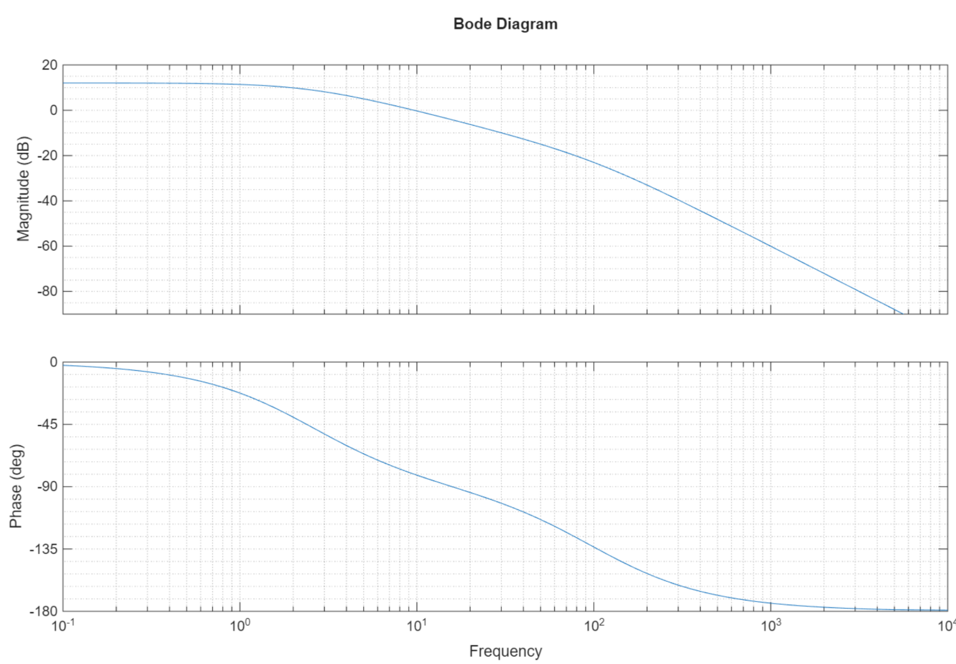
- When reducing the block diagram in part (b), you may treat blocks that are inactive in the given condition as if they do not exist.

Question 2 [20 marks]

You and some colleagues are working on identifying an unknown system. You start out by measuring the step response and get the following output:



Your colleague then measures the frequency response of the system and gets the following Bode plot:



It is suggested that the transfer function of this system is $P(s) = \frac{10}{s+2.5}$

a) Based only on the step response, show that $P(s)$ is a reasonable model for the system. [3 marks]

b) On the same sets of axes as the given Bode plot, sketch the approximate frequency response of $P(s)$. (A larger copy of the plot is provided on the next page.) [6 marks]

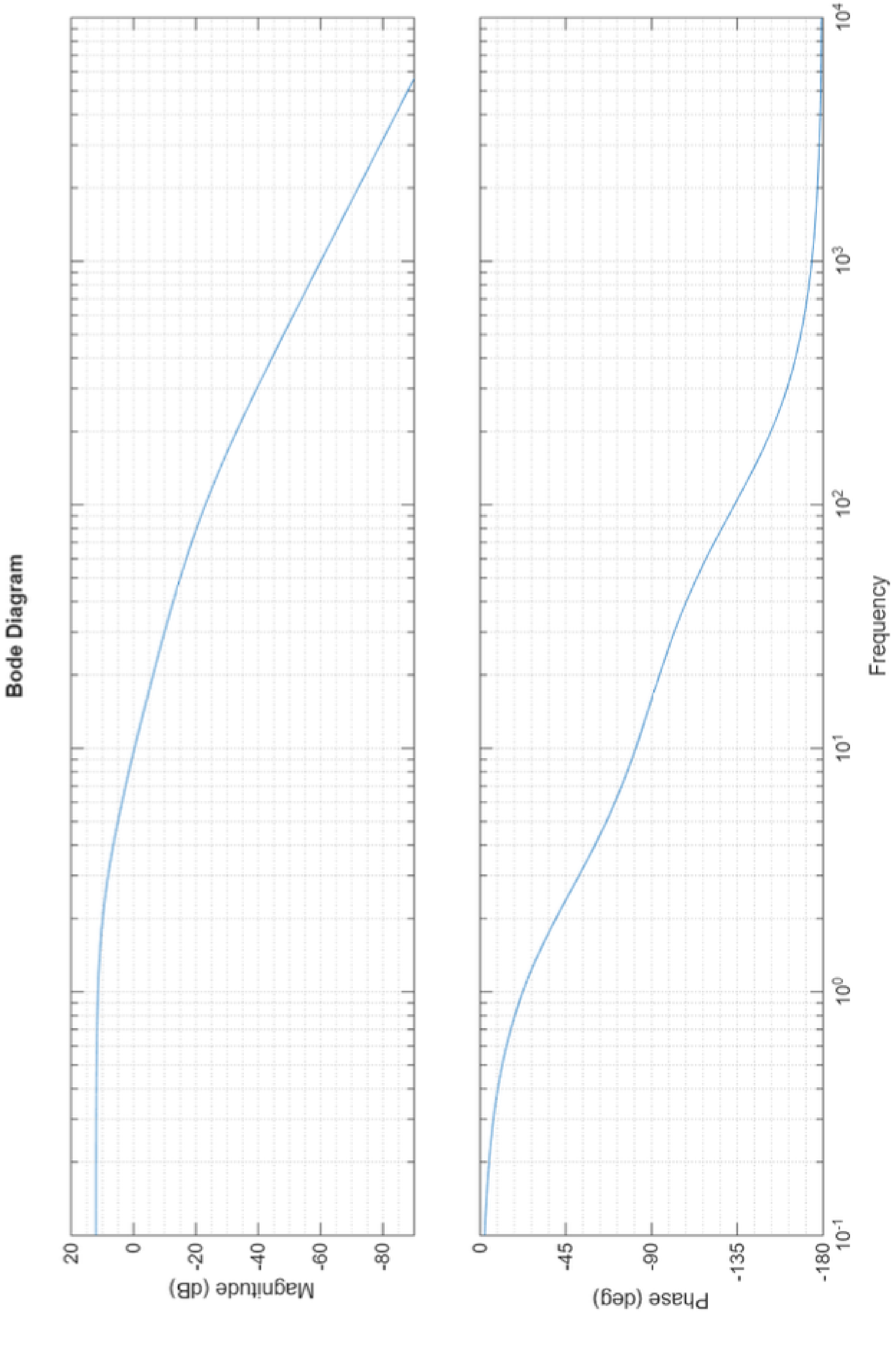
c) With the new information from the Bode plot, would you still consider $P(s)$ to be a valid model for the system,

   i. if the operating point of the system is around 2 rad/s? [2 marks]

   ii. if the operating point of the system is around 200 rad/s? [2 marks]

   Explain your answers.

d) Propose a higher-order model for the system that matches the order of the system as shown on the Bode plot. [3 marks]


Bode Diagram
Magnitude (dB)
20
0
-20
-40
-60
-80
Phase (deg)
0
-45
-90
-135
-180
10^-1
10^0
10^1
10^2
10^3
10^4
Frequency

Question 3 [15 marks]

Once upon a random Tuesday evening, Dr. Shield sat down to set a control test question. But sadly, her ADHD had other plans, and soon she was deep in a YouTube rabbit hole looking for footage of vaguely-remembered games from her childhood in the 90s so she could prove to herself that they actually existed. Like Hover! (exclamation mark included), a game where you race around various trippy maze-like environments in a little hovercraft and try to find all the flags before your opponent can.

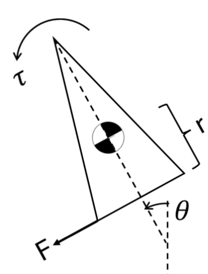
Luckily, this lead to an inspiring train of thought: "how would you code the controls of a simple hovercraft in a game?"

You want the vehicle to turn in a way that simulates the dynamics of a hovercraft, so it intuitively feels like one to the player. It should turn fast enough that it seems satisfyingly responsive, but not so fast that you overshoot the direction you want to go in. Which sounds like a modelling and gain selection problem! So this is now your test question.

Figure 2: Hover! A Windows 95 banger

Model:

A diagram of the model we will use to simulate the hovercraft is shown to the right. We will be focussing on the control of the steering. To turn, some portion of the hovercraft's forward thrust is redirected sideways, creating a torque that rotates it about its centre of mass. A small amount of air resistance, proportional to the hovercraft's rate of rotation, resists the turn.



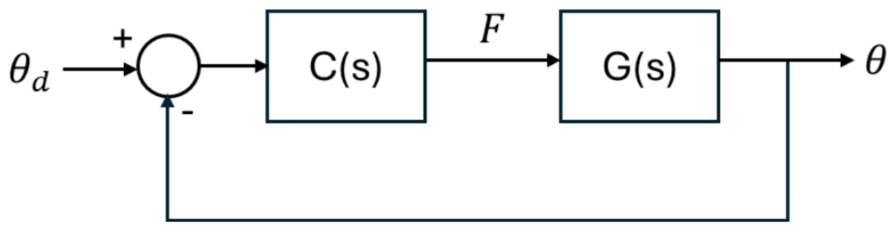
The change of the hovercraft's angle θ is therefore described by the differential equation

$$I\ddot{\theta} = rF - b\dot{\theta}$$

where I is the moment of inertia, r is the distance from the thruster to the center of mass, F is the lateral thrust, and b is the damping coefficient.

Control

We can think about the steering as a feedback control problem, where the player shifts the desired angle (θ_d) of the hovercraft by holding down the direction keys. The hovercraft ($G(s)$) will then receive an input thrust F that is related to the error between the actual angle θ and θ_d by the controller function $C(s)$. A block diagram representing this relationship is shown at the bottom of the page.



We need to design $C(s)$ so the hovercraft reaches the setpoint with perfect accuracy, as fast as possible without overshooting.

a) Derive a transfer function model relating the input F and output θ, assuming initial rest. [2 marks]

b) Is it possible to meet the 100% tracking accuracy specification with a proportional controller $C = K$? Explain your answer, and give a suggestion for a different type of controller if this one will not work. [3 marks]

c) Sketch a rough root locus for the problem, in terms of the system parameters r, I and b. [4 marks]
(Note: this is not a variation of parameters problem, we just mean that you must note any significant values on the root locus using the given symbols.)

d) What is the shortest possible settling time (to within 2% of the setpoint) for the system in terms of the system parameters, and under what damping conditions is it achieved? [4 marks]

e) Calculate the controller gain that would satisfy the specifications in terms of the other system parameters. [2 marks]